

**HOW TO USE THIS SET**

20 questions, mixed difficulty, 5 case-based. Attempt every question before checking the answer key on the last page.

4 Easy

10 Medium

6 Hard

1.**EASY**

Which zone of the lip has no hair follicles, sweat glands, or sebaceous glands, but is red due to prominent superficial vessels?

- A. Outer surface (skin)
- B. Inner (mucosal) surface
- C. Orbicularis oris
- D. None — all three zones are identical
- E. Vermilion border

2.**EASY**

Which papilla type is the smallest and most numerous on the tongue, and lacks taste buds entirely?

- A. Vallate
- B. Foliate
- C. Filiform
- D. Circumvallate
- E. Fungiform

3.**EASY**

Which major salivary gland is purely serous?

- A. Minor labial glands
- B. Sublingual
- C. Submandibular
- D. All are mixed
- E. Parotid

4.**EASY**

Which duct segment is identified by basal striations (vertical lines) in simple columnar epithelium?

- A. Striated duct
- B. Intercalated duct
- C. Main excretory duct only
- D. Intralobular gland acinus
- E. Interlobular duct

5.**MEDIUM****CASE**

A biopsy of oral mucosa shows stratified squamous epithelium resting directly on bone at low power, with minor salivary glands present in the submucosa above it.

Which region is this?

- A. Hard palate
- B. Floor of the mouth
- C. Cheek
- D. Soft palate
- E. Vermilion border

6.

MEDIUM

Circumvallate papillae are best identified by which combination of features?

- A. Small, conical, keratinized, no taste buds
- B. Large, sunk below the surface, ringed by a groove, with taste buds on the lateral walls and von Ebner gland ducts emptying into the groove
- C. Mushroom-shaped with a broad vascular core, scattered singly
- D. Absent from the human tongue
- E. Present only on the ventral tongue surface

7.

MEDIUM

CASE

An unfamiliar salivary gland field shows uniformly dark, granular, pyramidal cells forming round acini with narrow lumina; no pale tubules are seen anywhere.

Which gland is this, and what is the corollary for its duct system?

- A. Sublingual; short ill-defined intercalated ducts
- B. Submandibular; long striated and short intercalated ducts
- C. A minor mucous gland; no striated ducts present
- D. Parotid; long intercalated ducts and short striated ducts
- E. Sublingual; long striated ducts predominate

8.

MEDIUM

Which taste modality is best appreciated at the circumvallate papillae, at the back of the tongue?

- A. Umami exclusively, per this lecture
- B. Bitter
- C. Sour
- D. Sweet
- E. Salt

9.

MEDIUM

The submandibular gland is classified as which type of secretory gland, with which cell composition?

- A. Purely serous, compound acinar
- B. Compound tubulo-acinar, serous-predominant with mucous acini capped by serous demilunes
- C. Purely serous, compound tubular
- D. Purely mucous, compound tubulo-acinar
- E. Mixed with no serous predominance, exactly 50/50

10.

MEDIUM

Fungiform papillae are best distinguished from filiform papillae by which feature?

- A. Fungiform papillae are sunk below the surface in a groove
- B. Filiform papillae contain more taste buds than fungiform papillae
- C. Fungiform papillae have a mushroom shape, a richly vascularised connective tissue core, and scattered taste buds associated with sweet and salty taste
- D. Fungiform papillae are the most numerous and lack taste buds
- E. Fungiform papillae are found only at the back of the tongue

11.

MEDIUM

Serous acinar cells are distinguished from mucous acinar cells by which feature, per this lecture's description?

- A. Serous cells are cuboidal to columnar and arranged as tubules
- B. Mucous cells are always darker on H&E than serous cells
- C. Serous cells are always larger and paler than mucous cells
- D. Serous cells are pyramidal, arranged as a spherical acinus with a narrow lumen, and have basophilic cytoplasm with protein secretory granules
- E. Serous cells produce mucous secretion exclusively

12.

MEDIUM

Myoepithelial cells associated with salivary gland acini function to:

- A. Form the basement membrane of the duct only
- B. Contract and squeeze secretion out of the acinus toward the duct system
- C. Secrete mucus in place of mucous cells
- D. Regenerate acinar cells after injury
- E. Produce the primary serous secretion

13.

MEDIUM

The tongue's structural identification relies on which distinguishing histologic feature not seen in any other GIT organ?

- A. Goblet cells scattered throughout the epithelium
- B. Simple columnar epithelium overlying lymphoid tissue
- C. Skeletal muscle bundles cut in three planes within one field, capped by squamous epithelium
- D. A single plane of smooth muscle
- E. A stellate lumen with glands in the submucosa

14.

MEDIUM

The dorsal surface of the tongue, posterior to the sulcus terminalis, contains which structure?

- A. Von Ebner glands exclusively
- B. Foliate papillae exclusively
- C. Fungiform papillae exclusively
- D. Lingual tonsils
- E. Circumvallate papillae only

15.

HARD

CASE

A field shows a salivary gland with predominantly dark serous acini, but with conspicuously long, prominent striated ducts throughout the section, and demilunes capping occasional mucous tubules.

Which gland does this most likely represent, and what distinguishes its duct pattern from the parotid?

- A. Parotid, since long striated ducts are its hallmark
- B. Parotid, since demilunes are exclusive to the parotid
- C. A minor salivary gland, since major glands never show demilunes
- D. Submandibular, since it has long striated ducts and short intercalated ducts, the reverse of the parotid's long intercalated/short striated pattern
- E. Sublingual, since it is purely serous

16.

HARD

A patient loses parasympathetic innervation to a specific salivary gland known for producing thin, watery, protein-rich saliva. Which gland's secretory output would be most affected, and why?

- A. Parotid, since it is purely serous and produces the watery, protein-rich secretion described
- B. Submandibular, since it produces only mucous secretion
- C. Minor labial glands, since they dominate total saliva volume
- D. None — all glands produce identical secretions
- E. Sublingual, since it is purely mucous

17.

HARD

CASE

A biopsy of an unknown oral mucosal region shows thick non-keratinized stratified squamous epithelium with minor salivary (labial-type) glands beneath, sebaceous glands present, but no hair follicles or sweat glands.

Which region does this most likely represent?

- A. Vermilion border
- B. Inner surface of the lip
- C. Hard palate
- D. Soft palate
- E. Outer surface of the lip (skin)

18.

HARD

Given the lecture's own percentage breakdown (parotid 25–30%, submandibular 70%, sublingual 5%), which statement correctly reflects how these figures should be used on an exam?

- A. The percentages refer to gland size, not saliva contribution
- B. Only the sublingual percentage is considered reliable
- C. The percentages should be recalculated to sum exactly to 100% before answering any question
- D. The figures should be quoted as given in the lecture, even though they do not sum precisely to 100%, since they reflect the lecturer's own stated proportions
- E. These figures are irrelevant to identifying an unknown gland

19.

HARD

CASE

A pathologist examines an unknown papilla and must decide, before going to high power, whether it is circumvallate or one of the other types.

What is the single most efficient first discriminating question to ask at low power, per the scan path described in this lecture?

- A. Does the papilla have a vascular connective tissue core?
- B. Does the papilla contain taste buds at all?
- C. Does the papilla project above the surface, or does it sit sunk below the surface in a groove — sunk indicates circumvallate
- D. Is the papilla located on the ventral or dorsal tongue surface?
- E. Is the papilla keratinized or non-keratinized?

20.

HARD

A gland biopsy shows serous demilunes capping mucous tubules, with serous cells clearly predominating overall. Applying the discriminator from this lecture, which gland is this, and is the flagged 'staining' discrepancy relevant here?

- A. Cannot be determined without knowing gland size
- B. Parotid gland, since demilunes are its defining feature
- C. Submandibular gland, since serous-predominant with demilunes matches its described composition; the staining flag applies mainly to distinguishing acinus colour in trichrome-stained slides, not to this H&E-based identification
- D. Sublingual gland; the staining discrepancy is central to identifying it
- E. A minor salivary gland, since major glands never show mixed acini

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	E	Vermilion border
2	C	Filiform
3	E	Parotid
4	A	Striated duct
5	A	Hard palate
6	B	Large, sunk below the surface, ringed by a groove, with taste buds on the lateral walls and von Ebner gland ducts emptying into the groove
7	D	Parotid; long intercalated ducts and short striated ducts
8	B	Bitter
9	B	Compound tubulo-acinar, serous-predominant with mucous acini capped by serous demilunes
10	C	Fungiform papillae have a mushroom shape, a richly vascularised connective tissue core, and scattered taste buds associated with sweet and salty taste
11	D	Serous cells are pyramidal, arranged as a spherical acinus with a narrow lumen, and have basophilic cytoplasm with protein secretory granules
12	B	Contract and squeeze secretion out of the acinus toward the duct system
13	C	Skeletal muscle bundles cut in three planes within one field, capped by squamous epithelium
14	D	Lingual tonsils
15	D	Submandibular, since it has long striated ducts and short intercalated ducts, the reverse of the parotid's long intercalated/short striated pattern
16	A	Parotid, since it is purely serous and produces the watery, protein-rich secretion described
17	B	Inner surface of the lip
18	D	The figures should be quoted as given in the lecture, even though they do not sum precisely to 100%, since they reflect the lecturer's own stated proportions
19	C	Does the papilla project above the surface, or does it sit sunk below the surface in a groove — sunk indicates circumvallate
20	C	Submandibular gland, since serous-predominant with demilunes matches its described composition; the staining flag applies mainly to distinguishing acinus colour in trichrome-stained slides, not to this H&E-based identification

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Recalled and source questions are treated as evidence of what gets asked, not as a source of correct answers — verify against your lecture notes.